Chapter 4. Hierarchical Structure of Computer Systems

Hierarchical Structure

Von-Neumann Architecture

Bus Consept

One-to-one Connection
System Bus

Hierarchical Structure of Computer Systems

Just as in programming, large-scale problems are divided into modules and each module is designed separately, there is a similar structure in computer organization. Basically the computer hierarchy consists of 7 levels.

Level 6	**User Level**	Executable programs
Level 5	**High Level Languages**	C, Java etc.
Level 4	**Assembly Language**	Assembly code
Level 3	**System Software**	Like Operating System, Compilers
Level 2	**Machine Language**	Instruction set architecture
Level 1	**Control**	Hardware or micro programmed
Level 0	**Logic Level**	Gates, circuits, …

Hierarchical Structure of Computer Systems

Level 6 is the user level. It is the level where programs such as word processing programs, graphics, e-mail, and games that everyone uses are run.

Level 5 is the level of high-level languages. Includes languages such as C and Java. These languages must be translated into a language that the machine can understand using compilers or interpreters. Although at this level the user must know the data types and the operations they can perform on this data, they do not have to know how to perform these operations at lower levels.

Level 4 is the assembly language level. In assembly language, abbreviations are used that convey the meaning of the function to be performed (for example, using add for addition and sub for subtraction). After this level, the generated assembly codes can be converted one-to-one into machine codes.

Hierarchical Structure of Computer Systems

Level 3 is the system software level. It specifically concerns the responsibilities of the operating system. This level includes functions such as multiprogramming, memory organization, and synchronization between programs running together. These topics are within the scope of the operating systems course. Commands translated from assembly language into machine codes are passed to this level without being changed.

Level 2 is the machine level. At this level, there are machine instructions that can be recognized by a computer system with a certain architecture. Each processor has its own instruction set architecture. Programs written in machine language do not require a compiler, interpreter or converter and are executed directly on electronic circuits.

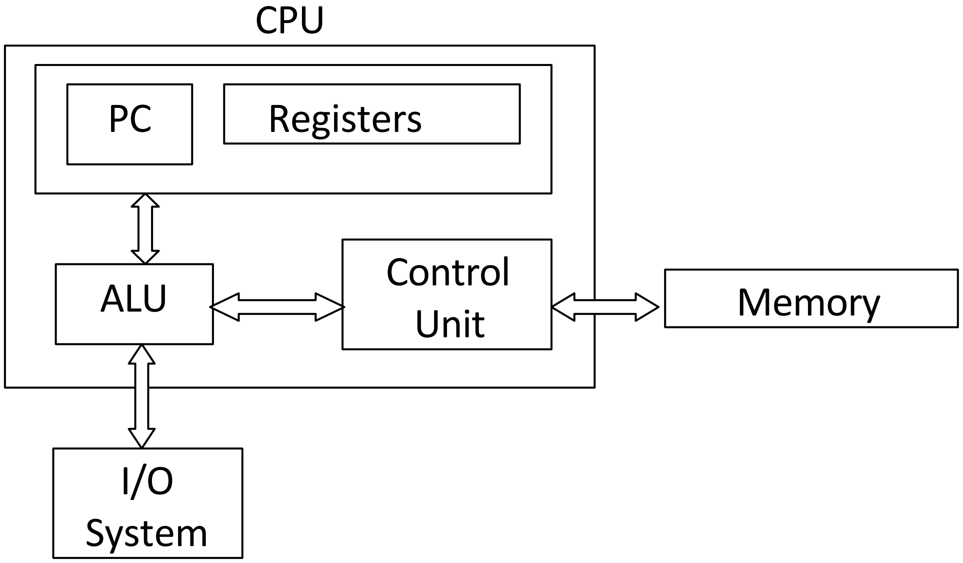
Hierarchical Structure of Computer Systems

Level 1 is the control level. The control unit interprets the machine commands coming from the next level and carries out the necessary action. The control unit can be implemented in two types: hardware (hardwired) or microprogrammed control. In hardware control, the control signals required to execute the commands are created directly through electronic mechanisms and these signals are transferred to the basic components that make up the computer. They are implemented (such as central processing unit, memory). In microprogrammed control, commands are executed through a microprogram. The microprogram is written in a low-level language that can be directly executed by hardware. Machine instructions at level 2 are routed to the microprogram, where each machine-level instruction is converted into multiple microcodes. Thanks to these microcodes, the necessary operations are carried out.

Level 0 is the logic level. At this level, there are basic physical components such as gates, electronic circuits, and buses.

Von-Neumann Architecture

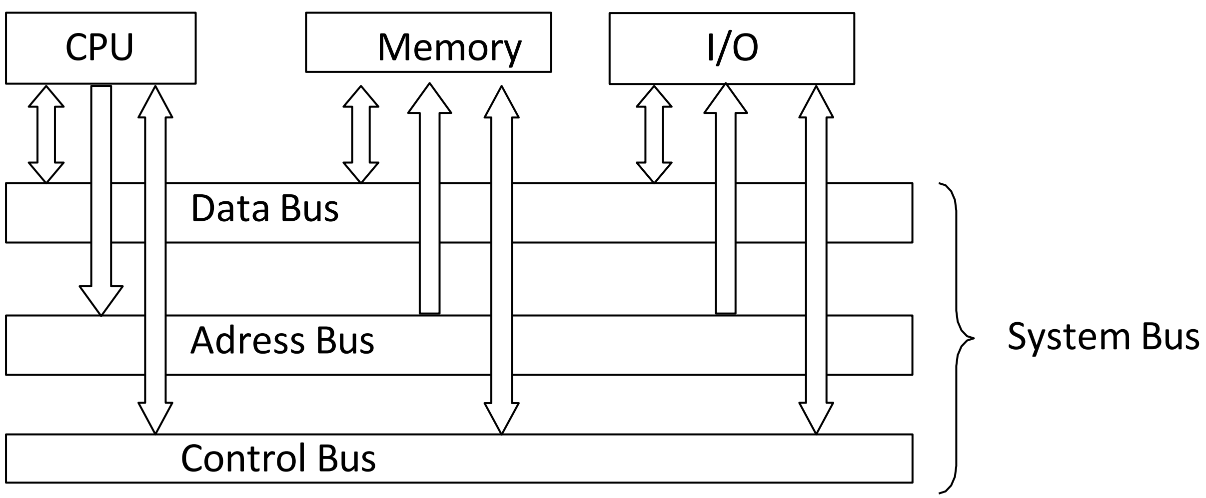
This architecture consists of 3 basic components; CPU (Central Processing Unit), memory and Input/Output devices. CPU consists of ALU (Arithmetic and Logic Unit), control unit, program counter (PC) and registers.



This architecture is capable of sequential command processing.
- The control unit receives the next command via the PC,
- The command is translated into a language that the ALU can understand,
- In case of operand, it is taken from memory or I/O units to CPU registers,
- The ALU executes the instruction and puts the result into registers, memory, or I/O units.

Von-Neumann Architecture

Today's computers are also based on this architecture. However, some expansions have been made. In particular, instead of limited connections between the CPU, main memory and I/O devices, all components are connected to each other via the system bus. Additionally, additional processor support for floating point numbers and the ability to perform parallel processing have also been added.



BUS Structure

There are generally three types of buses in a computer system;

- Adress bus
- Data bus
- Control bus

Buses can be organized in two ways;

- Single (Same lines are used for address and data)
- Multiple (Separate paths are used for address and data)

Generally, a multiple bus structure is used.

Connection Types

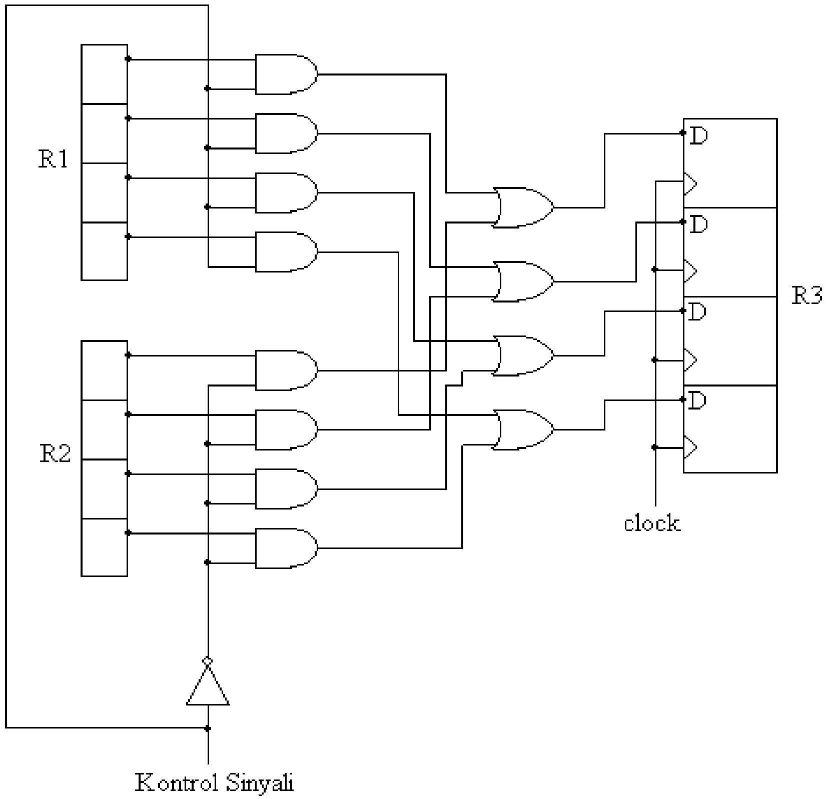
An important part of computer programs consists of data transfer between registers.

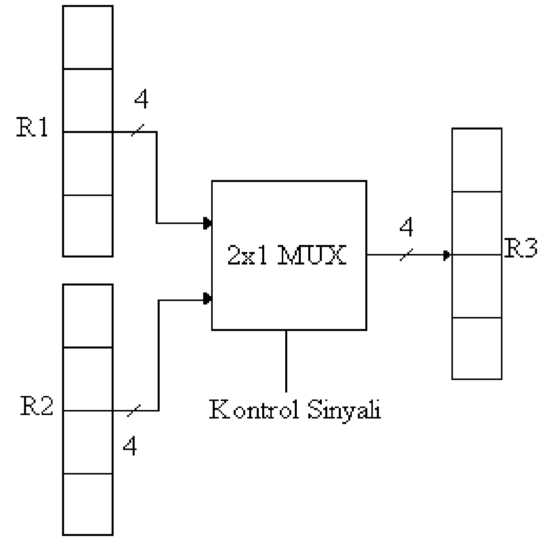
Data in one register can be transferred to several registers, or information can come to one register from one or more registers.

The connections between these components can be one-to-one. As the number of components increases, the number of connections will also increase, and a cable mess will occur. But data transfer is faster.

When a common bus structure is used, a simpler structure is obtained. Connections between all components occur through these pathways. The common bus is used with time sharing. While the content of a single register can be transferred to the common bus at the same time, data can be transferred to more than one register via the common bus.

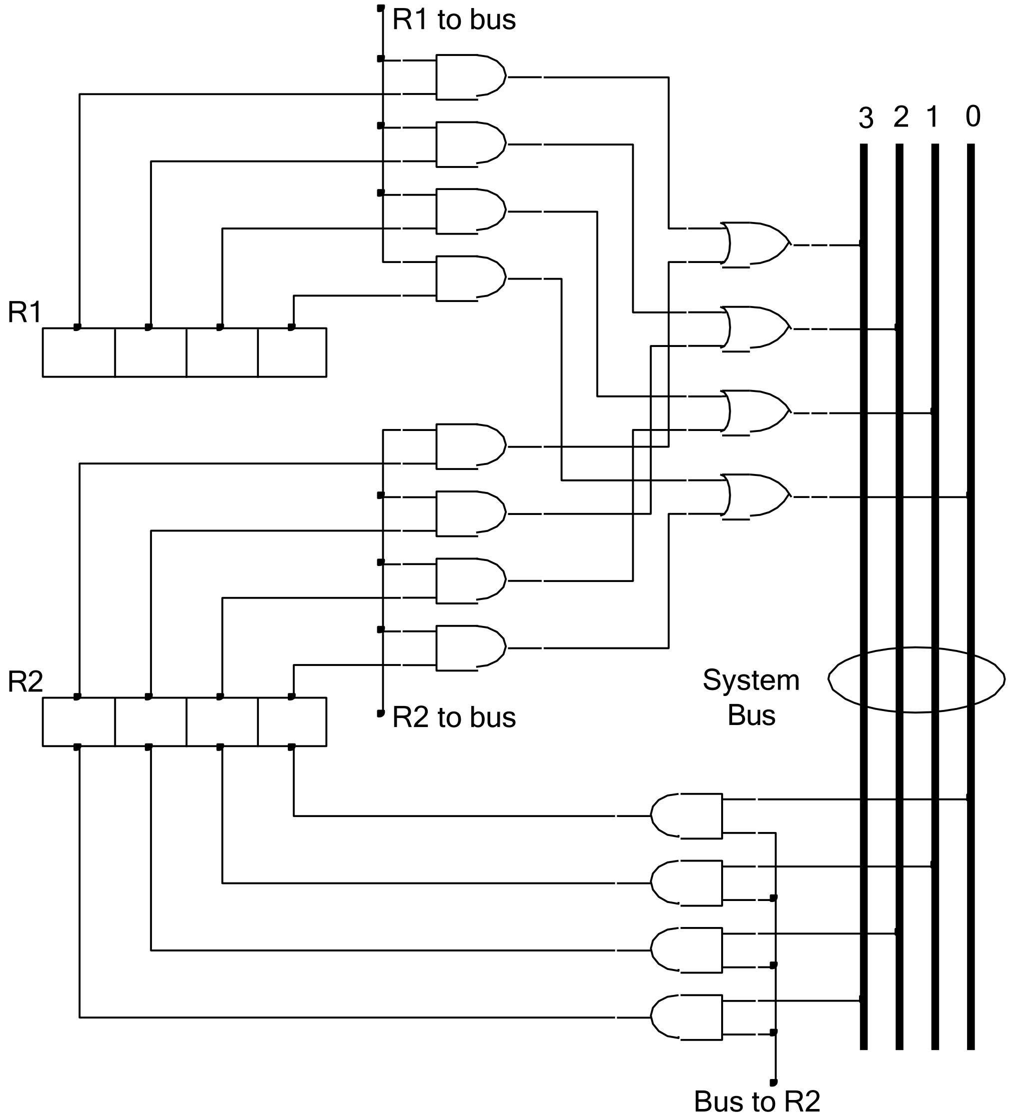
One-one Connection

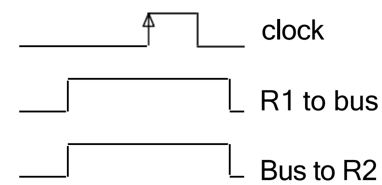




Depending on the value of the control signal, the content of R1 or R2 is transferred to R3.

System Bus Structure





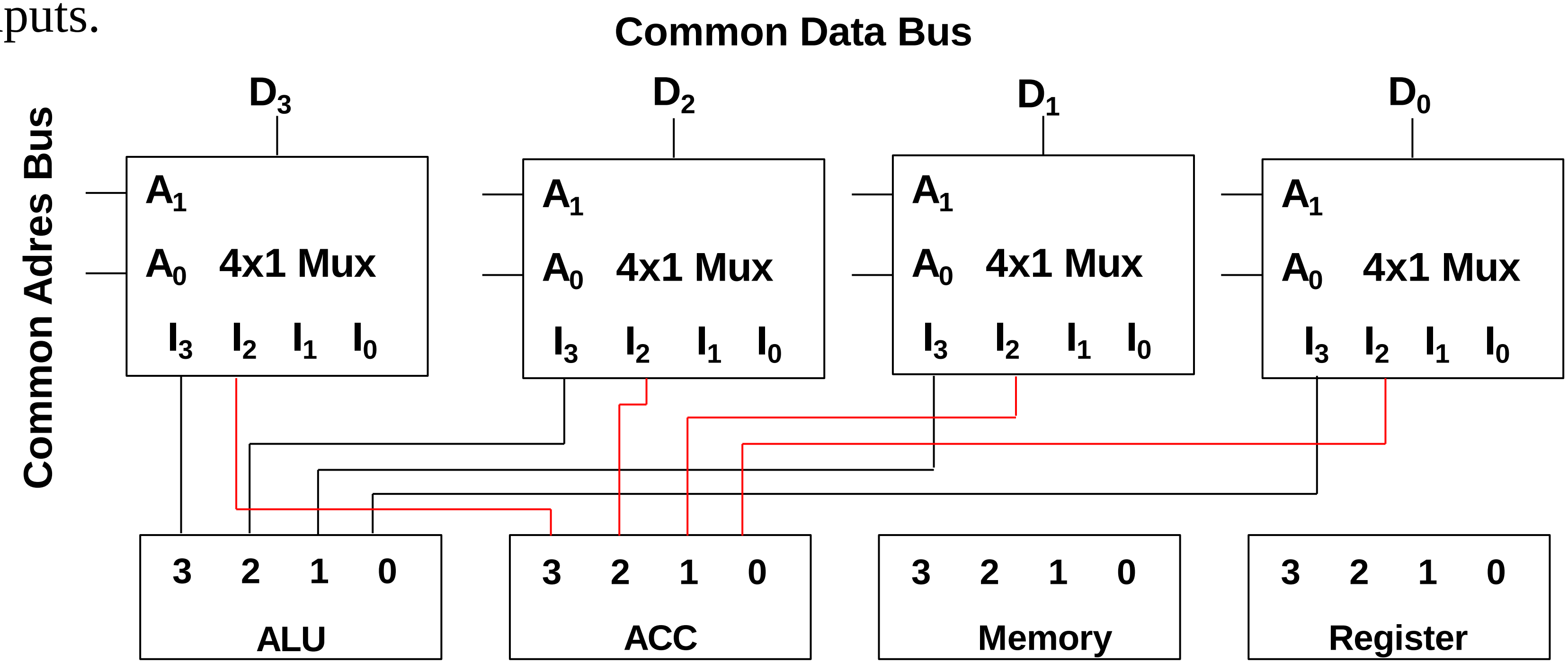
If you want to transfer data from R1 to R2, first the content of R1 must be transferred to the bus and then from the bus to R2.

System Bus Creation Methods

1. Using multiplexers (MUX),
2. Using three-state buffers (tristates)

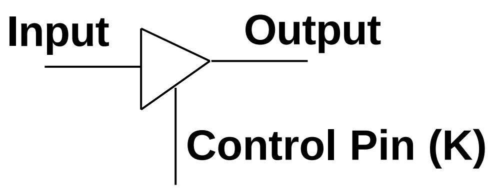
Example: Designing a 4-bit system bus using MUX.

Since our common bus will be 4-bit, 4 MUXes should be used. Considering that 4 components will be connected to this common bus, MUXs should have 4 inputs.



Tri-State Buffer

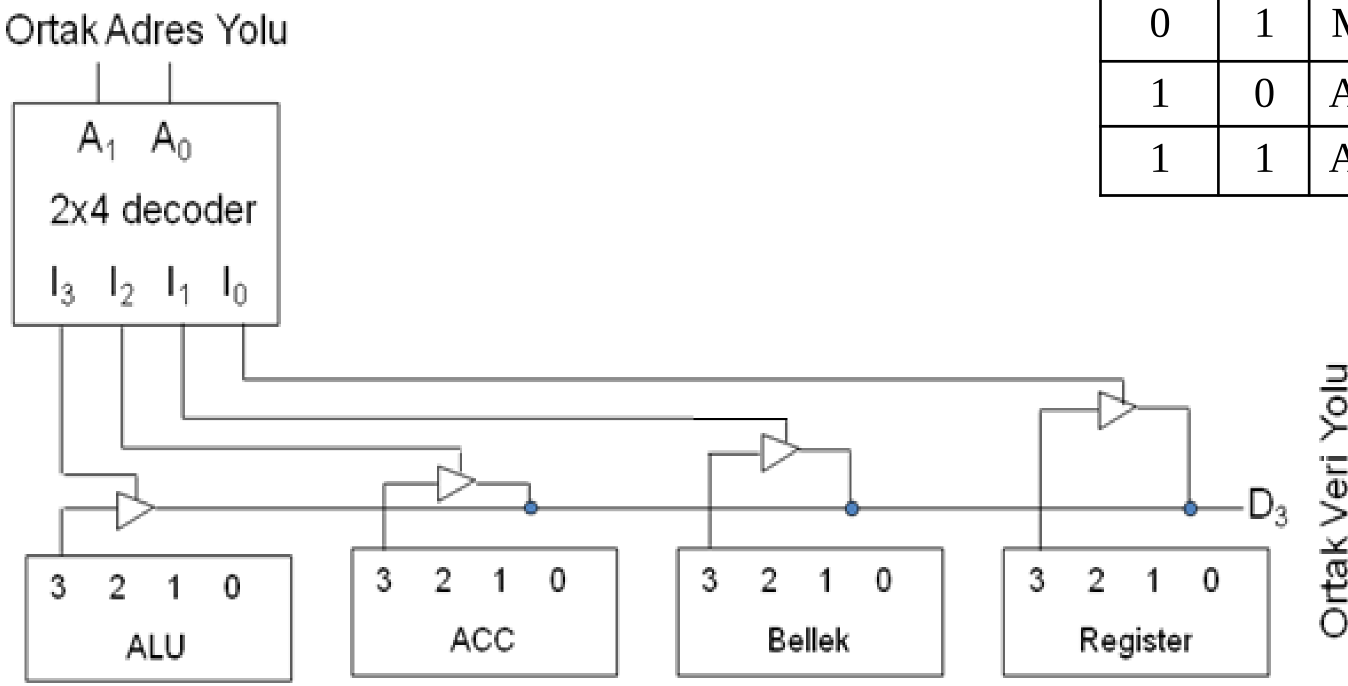
Decoder and three-state buffers can also be used to create a common path. The operation of this component is summarized below.



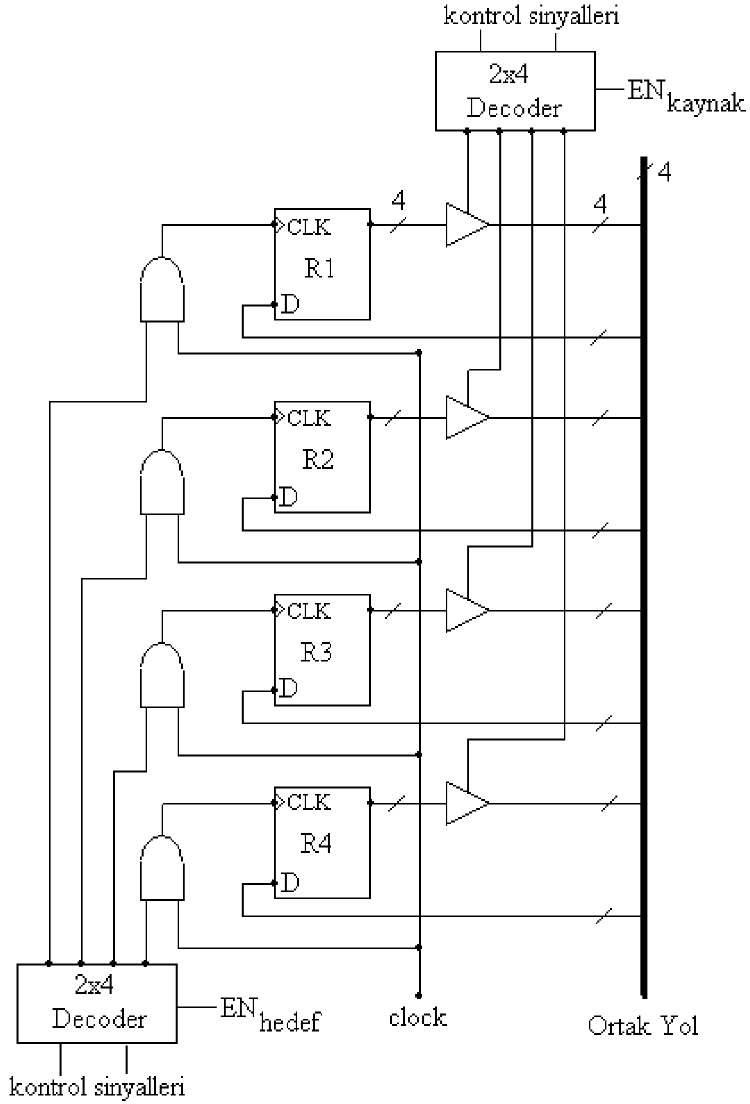
K = 1	Output $\leftarrow$ Input
K = 0	High impedance is observed at the output (insulation is provided).

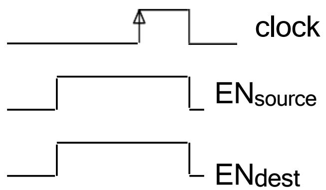
System Bus Design Using Tristates

A_1	A_0	Component that will use the bus
0	0	Register
0	1	Memory
1	0	ACC
1	1	ALU



A mechanism that allows the contents of the registers to be transferred to the bus and data received from the bus to the register.





Signals required for transmission